AI-HRMS ENTERPRISE PLATFORM

Complete Technical Architecture & System Specification

Comprehensive Engineering Deep-Dive: System Architecture, Relational Data Models, AI Engine Algorithms, Security Framework, REST APIs, and Cloud Infrastructure

Document Type:	Complete Engineering Specification	**Version:**	2.4.0 (Production)
Author:	AI-HRMS Core Engineering Team	**Status:**	Active / Verified Live
Target Database:	Supabase PostgreSQL (Cloud)	**Deployment:**	Vercel Serverless (Global Edge)
Frontend Framework:	React 18 + TypeScript + Vite	**Backend Framework:**	FastAPI (Python 3.11+)

1. Executive Summary & System Overview

AI-HRMS is a modern, cloud-native Human Resource Management System engineered for autonomous workforce governance, intelligent recruitment, precision GPS attendance verification, automated payroll computation, and predictive workforce intelligence. The platform unifies administrative operations with machine learning decision support.

The system is organized into a decoupled 3-tier distributed architecture ensuring low latency, high concurrency, and complete audit compliance:

Tier	Technology Stack	Core Responsibilities
Presentation Tier	React 18, TypeScript, Vite, Tailwind CSS, Lucide Icons, Recharts	Responsive web SPA, real-time geofencing geolocation, dashboard analytics, role-tailored viewports, dark/light theme engine.
Application & API Tier	FastAPI (Python 3.11+), Pydantic v2, Starlette, ReportLab PDF Engine	RESTful microservices, stateless JWT authentication, dual-layer RBAC, leave approval state machines, payroll algorithms, audit dispatchers.
Data & Intelligence Tier	Supabase PostgreSQL, SQLAlchemy 2.0 ORM, PyTorch / Scikit-learn, Google Gemini API	Relational persistence, connection pooling (port 6543/5432), immutable audit trail, AI resume parser, attrition prediction, and AI chat assistant.

2. Identity, Authentication & Role-Based Access Control (RBAC)

Security in AI-HRMS is anchored on a dual-layer RBAC architecture. Authentication utilizes cryptographically signed **JSON Web Tokens (JWT)** with HS256 encryption. Password storage strictly enforces standard **Bcrypt/Argon2** adaptive hashing algorithms.

Role Name	Key Permissions & Accessible Capabilities

SUPER_ADMIN	Full system access: global org settings, user creation, all department/employee records, geofencing coordinates, audit trail.
HR_MANAGER	Complete employee lifecycle governance, multi-tier leave approval, monthly attendance payroll processing, recruitment pipeline.
DEPARTMENT_MANAGER	Departmental staff oversight, 3+ day leave request approval/rejection, team performance radar, department analytics.
TEAM_LEADER	Direct team operational monitoring, up to 2-day leave request approvals, team attendance compliance reviews.
RECRUITER	Job vacancy authoring, AI resume screening, candidate competency scoring, interview pipeline scheduling.
EMPLOYEE	Self-service portal, GPS-verified check-in/out, leave applications, downloadable PDF payslips, AI growth recommendations.
CANDIDATE (Public)	Frictionless public application portal (No login required), open job vacancy exploration, PDF/DOCX resume upload with instant AI matching.

3. Relational Database Schema & Data Dictionary

The relational database schema is hosted on **Supabase PostgreSQL** and mapped via **SQLAlchemy ORM**. All foreign key relationships maintain referential integrity with cascading or set-null constraints where appropriate.

Table Name	Primary / Foreign Keys	Core Fields & Functional Description
users	PK: id	email (unique), hashed_password, full_name, role (Enum), is_active, last_login.
employees	PK: id FK: user_id, department_id, team_id	employee_code (unique), first_name, last_name, email, phone, designation, base_salary, employment_type, employment_status, date_of_joining, bank_details.
departments & teams	PK: id FK: manager_id, team_leader_id	name, code, description, manager_id (Department Head), team_leader_id (Team Lead).
attendances	PK: id FK: employee_id	date, punch_in, punch_out, punch_in_latitude, punch_in_longitude, distance_in_meters, verification_status (VERIFIED/FLAGGED/OUT_OF_BOUNDS), work_duration_hours.
leave_requests & leave_balances	PK: id FK: employee_id, approver_id	leave_type (CASUAL/SICK/EARNED/UNPAID), start_date, end_date, duration_days, reason, status (APPROVED/REJECTED/PENDING_TL/PENDING_MANAGER). Balance tracking per year.
jobs & candidates	PK: id FK: job_id, user_id (nullable)	Job requirements, salary range, candidate resume URL, extracted_skills, match_score, matching_skills, missing_skills, ai_summary, application status pipeline.
payrolls & payroll_items	PK: id FK: payroll_id, employee_id	month, year, working_days, present_days, lwp_days, basic_salary, allowances, lwp_deduction, pf_deduction, tax_deduction, net_salary, payslip_pdf_path.
salary_structures	PK: id FK: employee_id, department_id	3-tier priority compensation: basic_salary, hra, allowances, pf_percentage, tax_percentage, effective_from.
trainings & certifications	PK: id FK: course_id, employee_id	Enterprise learning modules, progress tracking, certificate upload, AI upskilling paths.
audit_logs	PK: id FK: user_id	action, module, record_id, details (JSON/Text), ip_address, timestamp (UTC). Immutable trail.

office_settings	PK: id	office_latitude, office_longitude, geofence_radius_meters, office_name, work_start_time, work_end_time.

4. Core Functional Modules & Engine Algorithms

4.1 Geofenced GPS Attendance Engine (Haversine Algorithm)

To prevent attendance spoofing and enforce physical presence verification, AI-HRMS executes the **Haversine Great-Circle Distance Formula** on every punch in/out attempt. Given the employee's browser GPS coordinates *(lat1, lon1)* and the configured office coordinates *(lat2, lon2)*:

```
d = 2 * R * arcsin( sqrt( sin²(Δlat / 2) + cos(lat1) * cos(lat2) * sin²(Δlon / 2) ) )
```

Where *R = 6,371,000 meters* (Earth's mean radius). If **d** ≤ **OfficeRadius (default 200m)**, the punch status is set to VERIFIED; otherwise it is recorded as OUT_OF_BOUNDS and flagged for HR review.

4.2 Intelligent Multi-Tier Leave Routing State Machine

Leave requests are evaluated through an automated state machine based on request duration:

- **1 Day Leave**: Instantly AUTO_APPROVED by the system, reducing administrative friction.
- **2 Days Leave**: Routed to the employee's designated TEAM_LEADER for operational approval.
- **3+ Days Leave**: Automatically escalated to the DEPARTMENT_MANAGER / HR_MANAGER.
- **Leave Balance Deduction**: Upon approval, the balance is deducted from Casual, Sick, or Earned leave pools; if depleted, it is marked as Leave Without Pay (LWP).

4.3 Autonomous AI Resume Parsing & Competency Matcher

The recruitment module processes uploaded PDF and Word documents via NLP keyword extraction and tokenization algorithms:

1. **Text Extraction & Entity Recognition**: Extracts contact details (Name, Email, Phone), education level, years of experience, and technical competencies.
2. **Weighted Competency Matching**: Computes the match percentage against the vacancy's required skills using set intersection and semantic relevance weighting.
3. **Frictionless Public Submission**: No applicant account or password required — candidates submit resumes freely at /careers.

4.4 Attendance-Based Payroll & Compensation Computation

Monthly payroll executes attendance-synchronized calculations:

- **3-Tier Salary Priority**: Employee Custom Rate > Designation Base Band > Departmental Minimum.
- **`Per-Day Rate`**: `PerDayRate = MonthlyBasicSalary / TotalWorkingDays`
- **`LWP Deduction`**: `LWPDeduction = PerDayRate * (WorkingDays - PresentDays - ApprovedLeaveDays)`
- **Statutory PF & Tax**: Provident Fund (12% of Basic) + Progressive Tax Withholding.
- **Automated PDF Payslip**: ReportLab generates a branded payslip PDF stored in Supabase/Local Storage for instant employee download.

4.5 AI Workforce Intelligence & Predictive Analytics

The predictive engine continuously analyzes attendance regularity, leave patterns, tenure, and compensation competitiveness to calculate:

- **Performance Prediction Index**: Multi-factor score forecasting high-impact contributors.
- **Attrition Risk Radar**: Early-warning retention flags (LOW, MEDIUM, HIGH) with identified risk and protective factors.

- **Skill Competency Matrix**: Visualizes team skill density and suggests targeted LMS course enrollments.

5. Comprehensive REST API Specifications

All endpoints reside under the /api route prefix. Standard HTTP status codes (200, 201, 400, 401, 403, 404, 500) and structured Pydantic schemas are enforced.

Module	Method & Path	Auth & Roles	Description
Auth	POST /api/auth/login	Public	Authenticates email/password; issues JWT bearer token.
Auth	GET /api/auth/me	Bearer (Any)	Returns current authenticated user profile and permissions.
Employees	GET /api/employees POST /api/employees	Super Admin, HR	Lists staff directory or registers new employee record.
Departments	GET /api/departments POST /api/departments	Super Admin, HR	Lists or creates organizational departments and sub-teams.
Attendance	POST /api/attendance/punch-in POST /api/attendance/punch-out	Employee	Records GPS timestamp and validates distance against geofence.
Leaves	GET /api/leaves POST /api/leaves/apply	Employee, HR, TL, Mgr	Applies for leave or triggers tiered approval state changes.
Recruitment	GET /api/recruitment/jobs POST /api/resumes/upload	Public (No Auth Required)	Public career vacancies and friction-free AI resume upload.
Payroll	POST /api/payroll/generate GET /api/payroll/payslip/{id}	Super Admin, HR, Employee	Generates monthly payroll cycle and renders PDF payslip.
Intelligence	GET /api/workforce/overview GET /api/workforce/attrition	Super Admin, HR, Manager	Retrieves workforce predictive scores and attrition radar.
Reports	GET /api/reports/{type}/export	Super Admin, HR	Streams authenticated CSV reports for Attendance, Payroll, Talent, Audit.

6. Deployment, Cloud Infrastructure & Production Security

AI-HRMS is deployed on a modern serverless cloud architecture with continuous automated integration (CI/CD):

- **Frontend Hosting**: Vercel Edge Network with worldwide HTTP/3 caching and Vite route-level code splitting.
- **Backend Microservices**: Vercel Python Serverless Runtime hosting FastAPI with auto-scaling instances.
- **Database Infrastructure**: Supabase Managed PostgreSQL with pooled transactions (port 6543) and direct connection (port 5432).
- **Immutable Security Audit Trail**: Every administrative action, department alteration, payroll run, and login attempt is written to the audit_logs table with IP address and UTC timestamp.
- **Automated Data Exports**: End-to-end authenticated CSV streaming directly from the Executive Reports center.

Document Approved By:	Aditya Kendre (Super Admin)	**Verification Date:**	2026-09-18
System Repository:	github.com/ADITYAKENDRE3254/ai-powered-hrms	**Live API Endpoint:**	backend-adityakendre3254.vercel.app